

Codebase Start Here

Orientation, reading path, setup, and the system mental model.

Eurskem AI · Agentic Workflow Platform

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Derived from the live repository

Source of truth. Inventories and symbol tables are generated from the current checkout. Narrative explains observed structures and does not replace executable contracts.

What this system is

Eurskem AI is a multi-user agentic workflow platform. Users author typed workflow graphs, validate them before execution, run them through a durable asynchronous runtime, observe live progress, pause for human decisions, and produce traceable research or business artifacts. The React application exposes Workflow Studio, Knowledge, evaluation, cost, chat, and operations surfaces over a FastAPI backend.

First-day reading path

- 1. Read this volume and the system architecture.
- 2. Follow one browser request through the frontend-backend map and API reference.
- 3. Learn workflow contracts and the runtime lifecycle.
- 4. Use volumes 17-23 as lookup references while changing code.

Repository map

- `app/api` : authenticated HTTP boundaries.
- `app/runtime` : workflow schema, validation, compilation, execution, events, and HITL.
- `app/nodes` : typed capabilities registered into the node manifest.
- `app/workflow` : durable workflow, conversation, run, file, and history stores.
- `app/llm` , `app/retrieval` , `app/rag` , `app/knowledge` : AI and grounded retrieval.
- `ui/src` : React SPA, API client, modes, components, and client state.
- `workflows` : executable and reference YAML graphs.

620 First-party files	3052 Python symbols	4335 Frontend symbols	171 HTTP routes
56 Node types	536 Workflow YAML	60 Test files	

Core invariants

- Workflows pass schema and preflight checks before provider tokens are consumed.
- Node outputs are validated against declared contracts.
- Security scope is resolved server-side and cannot be widened by caller metadata.
- Durable run records live outside process memory.
- External writes use authorization, policy, and idempotency controls.

File	Responsibility
<code>README.md</code>	<h1 align="center">Eurskem AI</h1> <p align="center"> An agentic workflow platform for high-stakes work.
 Typed workflows compiled to LangGraph,

File	Responsibility
	a zero-token preflight, a strict evidence lifecycle, and five canonical surfaces over one Workflow model.
app/main.py	FastAPI application entry point for Eurskem AI — Agentic Workflow Platform.
app/config.py	Config module.
app/runtime/schema.py	Schema module.
app/runtime/compiler.py	Compile a WorkflowSpec into a runnable LangGraph StateGraph.
app/runtime/preflight.py	Zero-token workflow validation.
app/nodes/base.py	Base contract for every node type in the platform.
app/api/runs.py	Workflow run history, inspection, pause/resume, retry, and deletion APIs.
ui/src/App.tsx	On a fresh page load the in-memory identity is empty, but the HttpOnly
ui/src/api/client.ts	---- auth identity (in-memory) ----